CLASS 6 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A picture 30 cm by 20 cm is mounted on a frame of width 3 cm all around. The area of the frame alone (cm^2) is: 1

(a) 348 **(b)** 336 **(c)** 339 **(d)** 333

2. A rod 6 m long is cut into pieces of $\frac{3}{4}$ m each. The number of pieces is: 1

(a) 10 **(b)** 8 **(c)** 9 **(d)** 7

3. Two circles of equal radius 4 cm touch externally. The distance from the outer end of one to the outer end of the other along the line of centres (cm): 1

(a) 20 **(b)** 24 **(c)** 12 **(d)** 16

4. The sum of the angles of rotation of a square and a regular hexagon (in degrees) is: 1

(a) 160 **(b)** 180 **(c)** 140 **(d)** 150

5. Two integers add to -5 and one of them is 12. The other is: 1

(a) -17 **(b)** -7 **(c)** -19 **(d)** -15

6. A wire in the shape of a square of side 20 cm is straightened and bent into a rectangle of length 26 cm. The breadth (cm) is: 1

(a) 8 **(b)** 6 **(c)** 26 **(d)** 14

7. If $\frac{3}{5}$ of a number is 12, then $\frac{1}{4}$ of the number is: 1

(a) 20 **(b)** 12 **(c)** 5 **(d)** 1

8. In rectangle ABCD, AC = BD = 8 cm. If O is the intersection, the sum OA + OB + OC + OD (cm) is: 1

(a) 16 **(b)** 12 **(c)** 24 **(d)** 20

9. A figure has rotational symmetry of order 3 and one of its lines of symmetry. The total number of lines of symmetry it must have is: 1

(a) 2 **(b)** 5 **(c)** 4 **(d)** 3

10. A trader loses ₹150 on day 1, gains ₹240 on day 2 and loses ₹120 on day 3. The net result in ₹ is: 1

(a) -32 **(b)** -20 **(c)** -30 **(d)** -28

11. Tiling a rectangular floor of length 9 m at ₹150 per m^2 costs ₹6750. Its breadth (m) is: 1

(a) 14 **(b)** 45 **(c)** 10 **(d)** 5

12. Which is the smallest of $\frac{5}{7}$, $\frac{7}{10}$, $\frac{3}{4}$, $\frac{8}{11}$? 1

(a) $\frac{8}{11}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{7}{10}$ **(d)** $\frac{5}{7}$

13. **Assertion (A):** A square has 4 lines of symmetry. 1
Reason (R): A regular polygon with n sides has n lines of symmetry.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** $-7 < -3$. 1
Reason (R): On a number line, a number lying to the left of another number is smaller.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The area of a square of side 7 cm is 49 cm^2. 1
Reason (R): Area is measured in square units.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A rectangular field is 40 m by 30 m. Find the cost of levelling the field at ₹5 per m^2 and fencing at ₹12 per metre. 2

17. Find the fraction which must be added to $\frac{2}{7}$ to get $\frac{5}{6}$. 2

18. A rectangle has diagonal intersection O with $\angle AOB = 35^\circ$. Find all four angles at O. 2

19. A figure fits onto itself after turns of 40° and 120° but not after any smaller turn than 40°. Find its order of symmetry and show that 120° is a multiple of 40°. 2

20. Find the missing number ?: $-9 + ? = 5$. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Two concentric circles have radii 4 cm and 9 cm. Find the width of the ring, the diameter of the inner circle and the length of the longest chord of the outer circle. 3

22. A regular polygon has 4 sides (4 is even). Of its 4 lines of symmetry, how many pass through two opposite vertices and how many pass through the midpoints of two opposite sides? 3

23. The sum of two integers is -4. One integer is 11. Find the other and their difference (larger minus smaller). 3

24. A floor 6 m by 4 m is to be covered with tiles each 30 cm by 20 cm. Find the number of tiles. 3

OR

Pipe A fills $\frac{1}{4}$ of a cistern in one hour and pipe B fills $\frac{1}{6}$ of it in one hour. What part of the cistern is filled in one hour when both work together? What part remains empty? 3

25. In rectangle ABCD, AB = 10 cm, BC = 6 cm. The diagonals meet at O. If M is the midpoint of AB and N is the midpoint of BC, find BM and BN, and the perimeter and area of the rectangle MBNO. 3

OR

A regular 6-gon is rotated about its centre by 180° anticlockwise and the figure looks unchanged. Verify this using the order of symmetry and find the number of rotations of the smallest angle that make 180°. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Case study. A pizza is cut into 8 equal slices. Kavita eats $\frac{3}{8}$ of the pizza and her brother Rohan eats $\frac{2}{8}$ of it. 5

(i) What fraction of the pizza did they eat together?

(ii) What fraction is left?

(iii) Who ate more?

(iv) How many slices are left, and what is the fraction left written with denominator 16?

27. Add $1\frac{1}{2}$, $2\frac{1}{3}$ and $3\frac{1}{6}$ and subtract $\frac{3}{4}$ from the sum. 5

OR

A designer draws 5 identical circles of radius 3 cm in a row, each touching the next. Find the total length of the row and the distance between the centres of the first and the last circle. How many circles of this size fit exactly in a strip 42 cm long? 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND